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(54) STRAWBERRY PLANT NAMED 'CIVL519'

(50) Latin Name: ***Fragaria* x *ananassa* Duchesne ex Rozier**
Varietal Denomination: **CIVL519**

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(52) **U.S. Cl.**
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See application file for complete search history.

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(57) **ABSTRACT**

This invention relates to a new and distinct variety of strawberry plant named 'CIVL519'. This new strawberry plant named 'CIVL519' is primarily adapted to the climate and growing conditions of Mediterranean and subtropical areas, and is primarily characterized by plants with compact leaves and visible fruits; easy to collect; fruits with regular and conical shape; easy to pollinate; excellent firmness consistency; and great taste with high sugar content and aroma.

4 Drawing Sheets

Latin name of the genus and species of the plant claimed: *Fragaria* x *ananassa* Duchesne ex Rozier.

Variety denomination: 'CIVL519'.

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from Community Plant Variety Rights Application No. 2021/1224, filed May 5, 2021.

BACKGROUND OF THE INVENTION

The present invention relates to a new and distinct strawberry variety named 'CIVL519'. The variety is botanically known as *Fragaria* x *ananassa* Duchesne ex Rozier.

The new strawberry 'CIVL519' is a product of a planned breeding program conducted by the inventors, Michelangelo Leis in San Giuseppe di Comacchio, Ferrara, Italy. The objective of the breeding program is to develop a new strawberry variety with high resistance to leaf and root disease. Fruits with regular and conical shape, bright colour, excellent consistence, good taste and aroma even high temperatures.

This new strawberry 'CIVL519' is a result of a controlled cross made by the inventors in 2013, in San Giuseppe di Comacchio, Ferrara, Italy. The female or seed parent is strawberry variety designated 9E1D-19 (unpatented CIV's selection). The male or pollen parent is strawberry variety designated 5F1I-1 (unpatented CIV's selection).

The new strawberry 'CIVL519' was discovered and selected by the inventor as a single flowering plant within the progeny of the stated cross in May 2016 in San Giuseppe di Comacchio, Ferrara, Italy (44° 45' North and 12° 11' East). After its selection, the new variety was asexually propagated by stolons in a nursery located in San Giuseppe di Comacchio, Ferrara, Italy. The new variety was extensively tested over the next several years in different European area with low chill condition. This propagation has demonstrated that the combination of characteristics as herein disclosed for the new cultivar are firmly fixed and retained through successive generations of asexual reproduction. The new cultivar reproduces true to type.

BRIEF SUMMARY OF THE INVENTION

'CIVL519' is primarily adapted to the climate and growing with low chill condition. This area provides the needed and correct temperatures for flower induction, and to produce a strong and vigorous plant, and to maintain fruit quality during full production.

The following traits have been repeatedly observed and are determined to be unique characteristics of 'CIVL519', which in combination distinguish this strawberry plant as a new and distinct variety:

1. plants with compact habit and visible fruits;
2. easy to collect;
3. fruits with regular and conical shape;
4. easy to pollinate;
5. excellent consistency; and
6. great taste with high sugar content and aroma.

Plants of the new strawberry variety 'CIVL519' differ from plants of the parents, 5F1I-1 and 9E1D-19 in the characteristics described in Table 1.

TABLE 1

Characteristic	'CIVL519'	5F1I-1	9E1D-19
Time of ripening	Very early to early	Medium	Early
Grow habit	Semi-upright	Upright	Semi-upright
Plant vigour	Weak	Medium	Medium to strong
Fruit size	Large to very large	Medium	Large
Firmness	Strong	Firm	Medium
Sweetness	Higher	Medium to high	Medium

Of the many commercial cultivars known to the present inventor, the most similar in comparison to the new strawberry variety 'CIVL519' is the strawberry variety VENTANA. Plants of the new strawberry variety 'CIVL519' differ from plants of strawberry variety VENTANA in the characteristics described in Table 2:

TABLE 2

Characteristic	'CIVL519'	VENTANA
Grow habit	Semi-upright	Semi-upright
Plant vigour	Weak	Medium
Fruit size	Large to very large	Large
Firmness	Strong	Medium
Sweetness	Higher	Medium
Time of beginning of flowering	Very early	Early
Time of beginning of ripening	Very early	Early

BRIEF DESCRIPTIONS OF THE PHOTOGRAPHS

The accompanying color photographs illustrate the overall appearance of typical specimens of the new strawberry variety 'CIVL519', at various stages of development as true as it is reasonably possible with color reproductions of this type. Color in the photographs may differ slightly from the color value cited in the botanical description which accurately describe the color of 'CIVL519'. The depicted plant and plant parts of the new strawberry variety 'CIVL519' were taken in Battipaglia, Salerno, Italy, and are approximately 6 to 9 months old.

FIG. **1** shows typical flowers of 'CIVL519';

FIG. **2** shows the leaves of 'CIVL519';

FIG. **3** shows typical fruits of 'CIVL519';

FIG. **4** shows plant of 'CIVL519'.

DETAILED BOTANICAL DESCRIPTION

'CIVL519' has not been observed under all possible environmental conditions. The characteristics of the new variety may vary in detail, depending upon variations in environmental factors, including weather (temperature, humidity and light intensity), day length, soil type and location.

The aforementioned photographs, together with the following observations, measurements and values describe the new strawberry variety 'CIVL519', unless otherwise noted, taken during March 2020 growing season in Battipaglia, Salerno, Italy. The observations, measurements and values were taken from plants of 'CIVL519' dug from Vivai Mazzoni Polska sp.z.o.o., Regon 240016883, kostomloty II 29, 21-509 Koden-Polonia, during October 2019 and planted approximately in the same month in CIV's test fields in Battipaglia, Salerno, Italy. Plants of the new strawberry variety 'CIVL519' were grown under conditions which closely approximate those generally used in commercial practice. The observed plants were one year old plants. The plants used in the production field are produced in a nursery in Vivai Mazzoni Polska sp.z.o.o., Regon 240016883, kostomloty II 29, 21-509 Koden-Polonia.

Yield observations and fruit quality characteristics are averaged from 2019 of data collected from the 2019 through 2021 growing seasons. Flower measurements and characteristics are from secondary flowers unless otherwise noted. Fruit characteristics and measurements are from secondary fruit unless otherwise noted.

Color references are made to The Royal Horticultural Society Colour Chart (R.H.S.), except where general colors of ordinary significance are used. Color values were taken under daylight conditions at approximately 3 pm in Battipaglia (SA) Italy. The approximate age of the observed plants is about 12 months.

The following tables 3-9 describe fruit, plant, stolon, foliage, fruiting truss, flower and pest/disease characteristics of the new strawberry 'CIVL519'.

TABLE 3

FRUIT CHARACTERISTICS	
Characteristic	'CIVL519'
Color of mature fruit	Red group - RHS 45A
Color of internal flesh	Red group - RHS 41D
Length (cm)	About 5.5 cm
Width (cm)	About 3.5 cm
Ratio length/width	Moderately longer
Calyx diameter (cm)	About 4.5 cm
Average weight (gm)	About 23 g
Achene color	Yellow-green group 150C
Number of achenes per cm^2 measured at the center of berry	About 7
Average weight of 1000 achenes (g)	About 0.67 g
Marketable yield (gm/plt)	Average 450 gm/plt
Size	Large
Predominant shape	Conical
Difference in shapes between primary and secondary fruit	Slight
Band without achenes	Absent
Unevenness of surface	Even or slightly uneven
Evenness of color	Strong
Glossiness	Strong
Insertion of achenes	Level with surface
Insertion of calyx	Level with fruit
Attitude of the calyx	Upward
Size of calyx in relation to fruit diameter	Slightly larger
Adherence of calyx	Strong
Firmness of skin	Firm
Firmness of flesh	Firm
Distribution of red color of the flesh	Uniform
Hollow center expression	Absent or small
Flavor	Good tase with good aroma and sugar content
Soluble solids (% brix)	About 9%
Time of first flowering	Very early
Time of first harvesting	Very early
Harvest period	From January until June in Battipaglia, Salerno, Italy
Type of bearing	Not remontant

TABLE 4

PLANT CHARACTERISTICS	
Characteristic	'CIVL519'
Height (cm)	About 10 to 20 cm
Spread (cm)	About 30 to 40 cm
Size	Medium
Habit	Semi-upright
Density	Medium
Vigor	Weak

TABLE 5

STOLON CHARACTERISTICS	
Characteristic	'CIVL519'
Average number per plant	About 15-20
Anthocyanin coloration	Red group - 53B
Anthocyanin intensity	Medium
Diameter at bract (mm)	About 3 mm
Pubescence	Dense

TABLE 6

FOLIAGE CHARACTERISTICS	
Characteristic	'CIVL519'
Foliage:	
Color of upper surface	Green group - 137A
Color of underside	Green group - 137C
Shape in cross section	Concave
Interveinal blistering	Medium
Glossiness	Medium
Number of leaflets	3
Terminal Leaflet:	
Length (cm)	About 6.1 cm
Width (cm)	About 6.5 cm
Length/width ratio	Equal
Serrations/leaf	Overlapping
Size	Medium
Shape of base	Rounded
Shape of teeth	Serrate to crenate
Petiole:	
Length (cm)	About 13 cm
Diameter (mm)	About 2.7 mm
Pubescence	Medium
Attitude of hairs	Horizontal
Stipule:	
Length (mm)	About 35 mm
Width (mm)	About 9 mm
Anthocyanin coloration	Weak
Color	Yellow-green group 145C

TABLE 7

FRUITING TRUSS CHARACTERISTICS	
Characteristic	'CIVL519'
Length (cm)	About 20-25 cm
Attitude	Erect
Position relative to foliage	Above
Pubescence	Strong
Antbocyanin intensity	Weak
Attitude at first pick	Horizontal

TABLE 8

FLOWER CHARACTERISTICS	
Characteristic	'CIVL519'
Petal color	
Mature (upper)	White group - 155D
Mature (lower)	White group - 155D
Petal shape	
Overall	Transverse elliptic
Apex	Rounded
Base	Rounded
Petal length (mm)	About 1 to 1.5 cm
Petal width (mm)	About 0.80 to 1.3 cm
Petal length/width ratio	Moderately longer
Number of petals/flower	About 5 to 7
Sepals color	
Mature (upper)	Green group - 137A
Mature (lower)	Green group - 138A
Sepal shape	
Overall	Narrow elliptic
Apex	Pointed
Base	Flat
Sepal length (mm)	From 1 to 1.5 cm
Sepal width (mm)	From 0.5 to 0.7 cm
Sepal length/width ratio	Longer
Number of sepals/flower	About 9 to 13
Corolla diameter (mm)	About 30 mm
Calyx diameter (mm)	About 30 mm
Size of calyx relative to corolla	Equal
Size of inner calyx relative to outer calyx	Smaller
Relative position of petals	Overlapping

TABLE 9

REPRODUCTIVE ORGANS	
Characteristic	'CIVL519'
Receptacle color	Yellow green group - 154B
Pollen color	Yellow orange group - 14A
Stamen	Present
Pollen amount	Much

TABLE 10

PEST AND DISEASE REACTIONS	
Characteristic	'CIVL519'
Powdery mildew (*Sphaerotheca macularis*)	Tollerant
Angular leaf spot (*Xanthomonas fragariae*)	Moderately susceptible
Botrytis fruit rot (*Botrytis cinerea*)	Very low sensibility
Fusarium wilt (*Fusarium oxysporum*)	Moderately susceptible
Anthracnose crown rot (*Colletotrichum fragariae*)	Moderately susceptible
Two-spotted spider mite (*Tetranychus urticae*)	Susceptible

I claim:

1. A new and distinct strawberry plant named 'CIVL519', as herein described and illustrated by the characteristics set forth above.

* * * * *

FIG. 1

FIG. 2

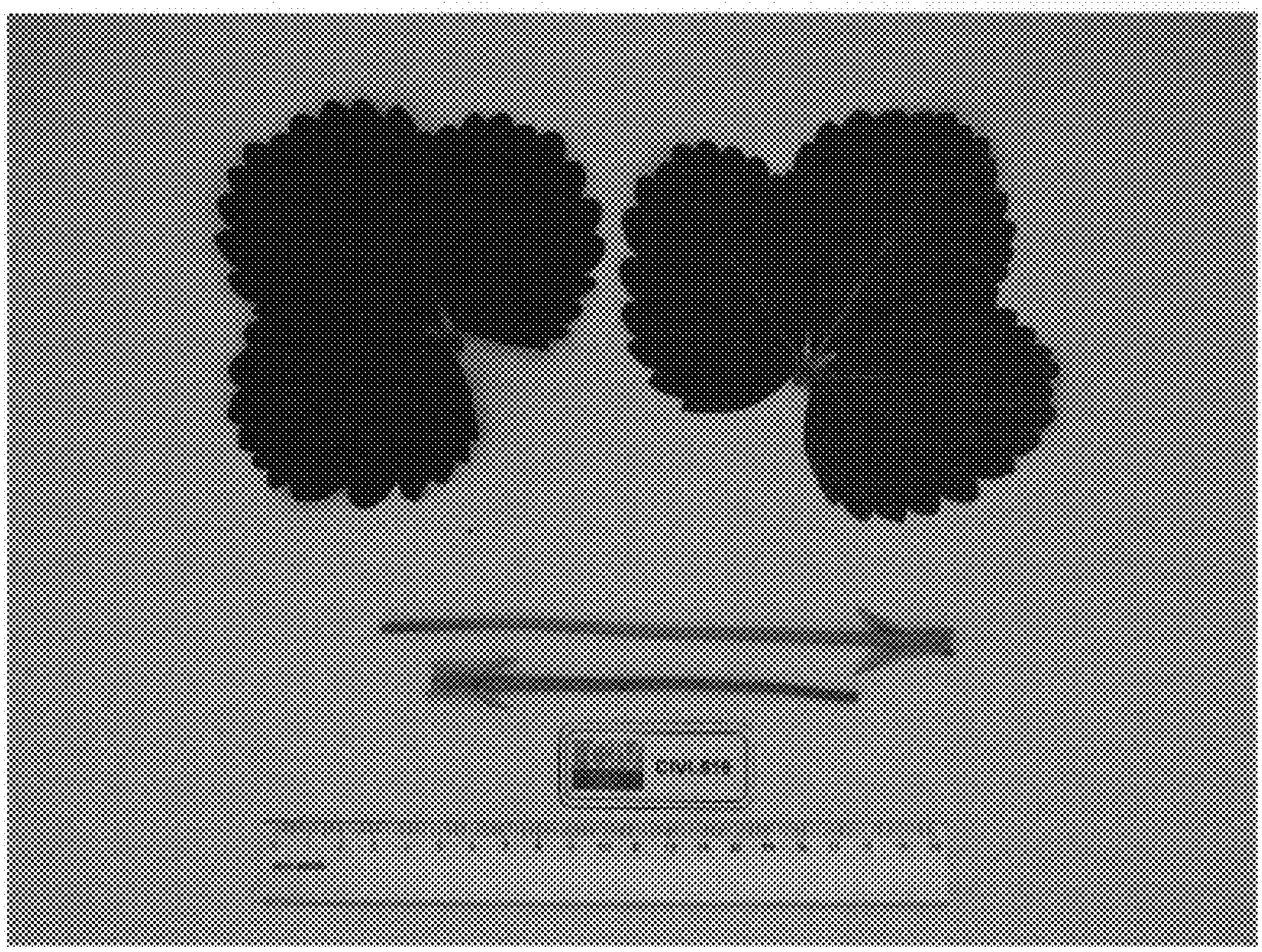

FIG. 3

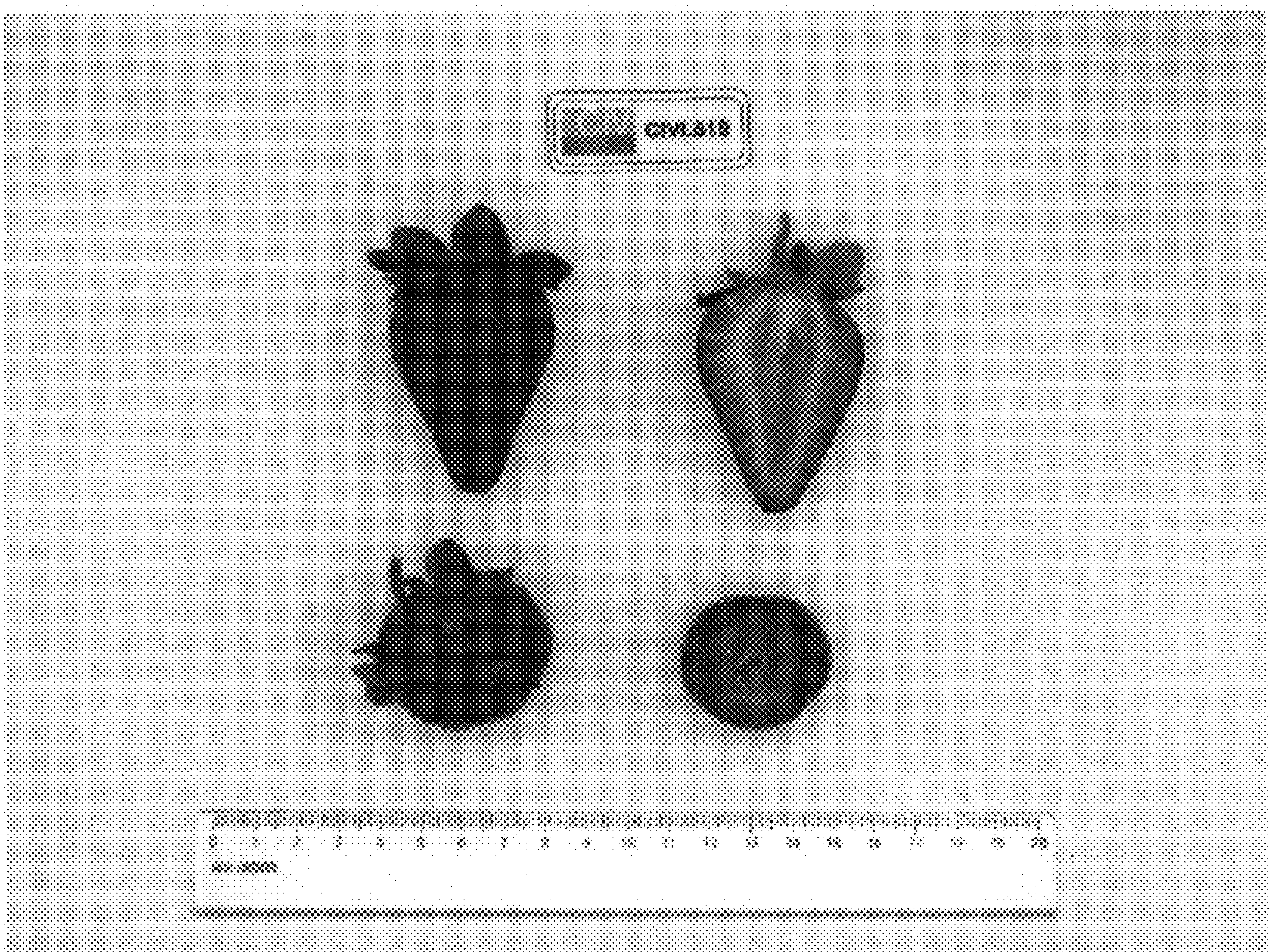

FIG. 4